
Thinking in Emojis

A Proposal for UROP/FYP/MComp Projects

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1 Motivation

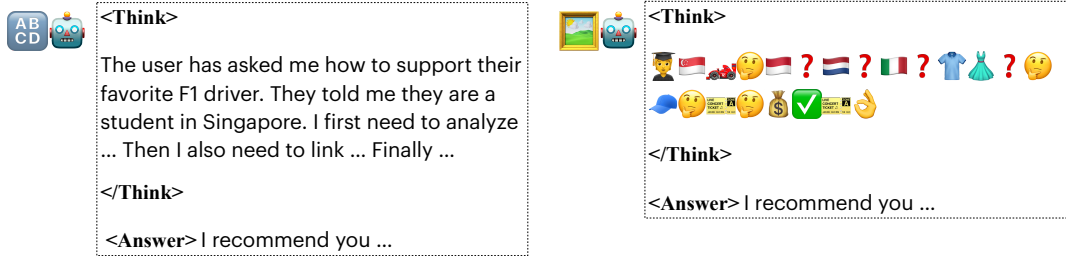


Figure 1: Comparison of thinking in English text (left) versus emojis (right).

In this project, we propose to innovate how large language models (LLMs) *think* by replacing lengthy text tokens with emoji-only strings (Figure 1). Emojis are widely used in communication to convey compressed meaning. Using emojis for internal reasoning offers the following advantages:

For Reasoning with LLMs: (1) Reasoning processes are often lengthy and costly [CVD24]. Emojis encode meaning more densely, improving token efficiency; (2) Reasoning across different languages can diverge in outcomes [LN24]. Unlike verbal languages, emojis are globally recognized (though interpretations may vary subtly), potentially enabling more consistent reasoning; (3) Prior work on unnatural language [DZF⁺25] and “Neuralese” [ADK17] shows that non-standard languages and representations can still be effective. Emojis provide a similar alternative, with the added benefit of being interpretable as a medium between natural language and latent embeddings.

For the Study of Emojis: The computational linguistics (CL) community has explored whether emojis can function as a language. Studies of emoji grammar suggest they are not suitable for sentence-level composition [CES19]. The ELCo dataset models emoji composition but remains at a surface level [YZM24]. Using emojis for reasoning could provide new insights for CL: if long emoji sequences are useful for models (even if not naturally interpretable by humans), can we analyze them to derive meaning and develop an emoji grammar?

This proposal is also available at <https://yisong.me/topics/ThinkInEmojis.pdf> and will remain online until the position is filled (we will switch to the stealth mode by then).

2 Approach

An envisioned project cycle consists of the following steps: **Step 1:** Construct bitexts between English reasoning and emoji reasoning; **Step 2:** Evaluate and improve emoji-based reasoning. A promising approach is alternating optimization, where we iteratively optimize both the generation of emoji reasoning and the model’s ability to interpret and use it for producing answers; **Step 3:** Parse and translate emoji reasoning. Once its performance matches that of English reasoning, this will demonstrate that the model can effectively *think* in emojis. At this stage, we will explore the potential grammar and constituency rules for emoji sequences [CES19, YZM24].

3 Organization

Who will I be working with? In this project, you will be working with WING PhD student Yisong Miao and WING lead Prof. Min-Yen Kan.

When is the starting date? January 2026 or earlier (e.g. October 2025).

What are the prerequisites? Strong natural language processing (NLP) experience, e.g., you have led a team to excel in the CS4248 / CS6101 project components (or equivalent).

What are the expected commitments and outcome? For UROP/FYP/MComp students, we expect sufficient time commitment to achieve excellence in this project. The expected outcome is a research publication, so applicants with academic inclinations are preferred.

How to get in touch? Please email Yisong and Prof. Min. We also copy Min’s message: “You *must* meet with me on Zoom or via email before deciding to take this project or any other project under my group. Please email me first to arrange a suitable timing. If you bid without consulting me, I will refuse your project bid. Read my notes on projects (<http://www.comp.nus.edu.sg/~kanmy/hyp.html>) and learn about the host organization, WING <http://wing.comp.nus.edu.sg/>, and read the requirements about entry to WING: <http://bit.ly/new-to-wing> (there’s a narrated video at <http://bit.ly/new-to-wing-yt> as well).” This project is primarily supervised by Yisong, a doctoral student under Min. Min will be overseeing the project, but not necessarily working directly with you.

References

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- [CVD24] Jeffrey Cheng and Benjamin Van Durme. Compressed chain of thought: Efficient reasoning through dense representations. *arXiv preprint arXiv:2412.13171*, 2024.
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- [YZM24] Zi Yun Yang, Ziqing Zhang, and Yisong Miao. The ELCo dataset: Bridging emoji and lexical composition. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15899–15909, Torino, Italia, May 2024. ELRA and ICCL.